Math 477 Homework 8

Spring 2026
Rutgers University

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Question 1:

Let X and Y have a joint density function:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{x^2y^2} & x \geq 1, y \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Let $U = XY$ and $V = X/y$. Then, $X = \sqrt{UV}$ and $Y = \sqrt{\frac{U}{V}}$. So

$$J = -\frac{1}{2v}.$$

Thus,

$$f_{X,Y}(x,y) = J \cdot \frac{1}{x^2y^2} = \frac{1}{2vu^2}.$$

(b)

$$\begin{aligned} f_U(u) &= \int_{1/u}^{u} \frac{1}{2u^2v}\, dv \\ &= \frac{\ln u}{u^2} \end{aligned}$$

and

$$\begin{aligned} f_V(u) &= \int_{v}^{\infty} \frac{1}{2u^2v}\, dv \\ &= \frac{1}{2v^2} \end{aligned}$$

Question 2:

We have

$$\begin{aligned} f_Y(y) &= \frac{e^{-y}}{y} \int_0^{\infty} e^{-x/y} \\ &= e^{-y}. \end{aligned}$$

Then,

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{e^{-x/y}e^{-y}}{ye^{-y}} \\ &= \frac{e^{-x/y}}{y}. \end{aligned}$$

Therefore,

$$\begin{aligned} E[X^2|Y=y] &= \int_0^\infty x^2 \cdot \frac{e^{-x/y}}{y} \\ &= y^2 \int_0^\infty u^2 \cdot e^{-u} \\ &= y^2 \cdot \Gamma(3) \\ &= 2y^2 \end{aligned}$$

Question 3:

For each person $i \in \{1, 2, \ldots, 1000\}$, let

$$I_i = \begin{cases} 1 & \text{if person } i \text{ recieves card number } i \\ 0 & otherwise \end{cases}.$$

So $E[I_i] = P(I_i = 1) = \frac{1}{1000}$. Then,

$$\begin{aligned} E\left[\sum_{i=1}^{1000} I_i\right] &= \sum_{i=1}^{1000} E[I_i] \\ &= 1. \end{aligned}$$